

ANDRES SANTIAGO DUCUARA VELASQUEZ

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EDUCATION

Masters

- **Master in Artificial Intelligence** **Universidad Sergio Arboleda**
M.Sc. in Artificial Intelligence April 2026

Graduate

- **Computer Science & Artificial Intelligence** **Universidad Sergio Arboleda**
Engineer in Computer Science and Artificial Intelligence April 2024

EXPERIENCE

National Ministry of Defense

January 2026 – December 2026

Artificial Intelligence Engineer

Bogotá, Colombia

- Provided technical and methodological support to the Public Security Forces in the planning, monitoring, and enhancement of institutional capabilities in science, technology, and innovation, ensuring alignment with current policies and strategic guidelines.
- Delivered specialized advisory services in strategic information analysis, funding strategy development, intellectual property protection, and technology transfer to support decision-making processes and the modernization of the Defense sector.

Universidad Sergio Arboleda

February 2025 – May 2026

Professor

Bogotá, COL

- In charge of the course "Algorithmic Thinking," which covers introductory engineering topics such as logic, problem-solving, and the fundamentals of programming.
- In charge of the course "Cybernetics and Intelligent Systems," which explores the core principles of cybernetics, the early developments of artificial intelligence with McCulloch-Pitts, and complex systems such as chaos theory.

Contraloría General de la República

August 2024 – December 2024

Data Engineer

Bogotá, COL

- Structuring and continuous improvement of analytical models through efficient automation of each stage of the ETL process.
- Implementation of solutions based on Azure and Databricks, using Python and SQL to optimize the processing and analysis of large data volumes.
- Design and deployment of interactive dashboards in Power BI for effective visualization of analytical results.

Unidad de Planeación Minero Energética - UPME

May 2024 – December 2024

Artificial Intelligence Engineer

Bogotá, COL

- Development of pilot projects and research of artificial intelligence solutions to optimize critical processes within the organization.
- Integration of artificial intelligence solutions.

Contraloría General de la República

November 2023 – December 2023

Data Analyst

Bogotá, COL

- Provision of services to support document management for the Information Unit of the Directorate of Information, Analysis, and Immediate Reaction (DIARI).
- Support in various projects of the entity related to OCR and web applications on Azure cloud architecture.

Universidad Sergio Arboleda

Mathematical talent's teacher

August 2023 – June 2024

Bogotá, COL

- Teaching courses related to technology, such as robotics, communication protocols, programming fundamentals, among others, aimed at children and adolescents.

Contraloría General de la República

Data Engineer

January 2022 – August 2022

Bogotá, COL

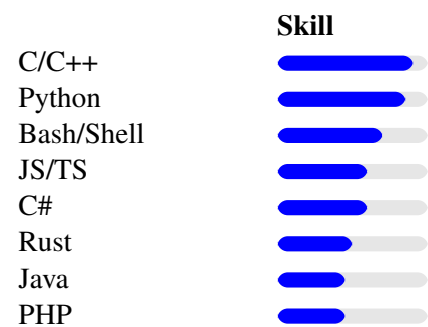
- Provision of management support services in the administrative assistance, handling of information and documentation of the activities of the DIARI. (Dirección de información, Análisis y Reacción Inmediata).
- Structuring public data from territorial entities at the central level and universities, implementing Azure services (Pipelines and Databricks).
- Development projects related to the use of OCR, web scraping, Python, or Azure applications.

CERTIFICATES

- Fundamentals of Accelerated Computing with CUDA C/C++ [NVIDIA DLI]
- NDG Linux Essentials [NDG Netdevgroup]
- Introduction to Packet tracer [CISCO Networking Academy]
- Programming Essentials in Python [EDG Python Institute]
- Programming Essentials in C++ [EDG C++ Institute]
- Database Design and Programming with SQL – Oracle [Oracle Academy]
- Programación en Java [Politécnico de Colombia]
- Programación en PHP [Politécnico de Colombia]

TECHNICAL STRENGTHS

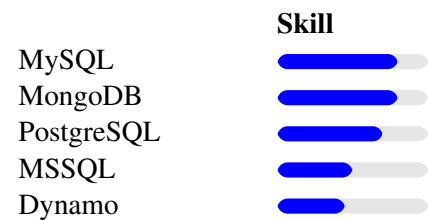
Computer Languages



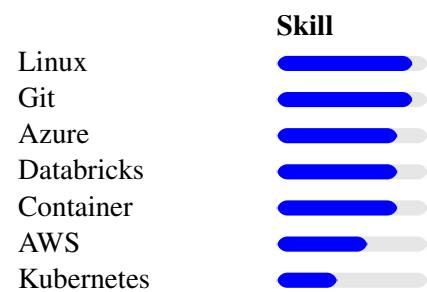
Frameworks



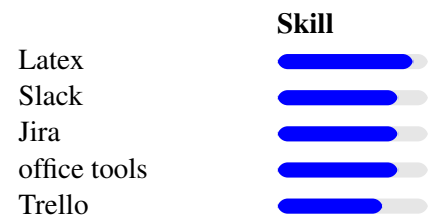
Databases



Cloud & DevOps skills



Productivity & collaboration suites



High performance & AI computing

- **ML:** TensorFlow, PyTorch, Keras, Scikit-learn
- **GPU Computing:** CUDA, Numba, OpenCL
- **NLP:** Hugging Face, Transformers, LangChain
- **Parallel Computing:** MPI, OpenMP, SYCL
- **Distributed Systems:** Apache Spark, Hadoop
- **Electronics:** Embedded Systems, IoT, Robotics, EDA